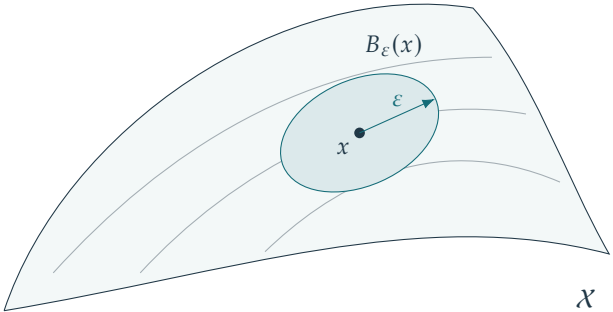

geodesic ball

$\dim \mathcal{X} = d$

$$\frac{1}{\mathrm{Vol}(B_\varepsilon(x))} \int_{B_\varepsilon(x)} f(y)\, \mathrm{d}\mu(y) = f(x) + \frac{\varepsilon^2}{2(d+2)} \Delta f(x) + o(\varepsilon^2)$$